RAK12059 WisBlock Liquid Level Sensor Datasheet

Overview

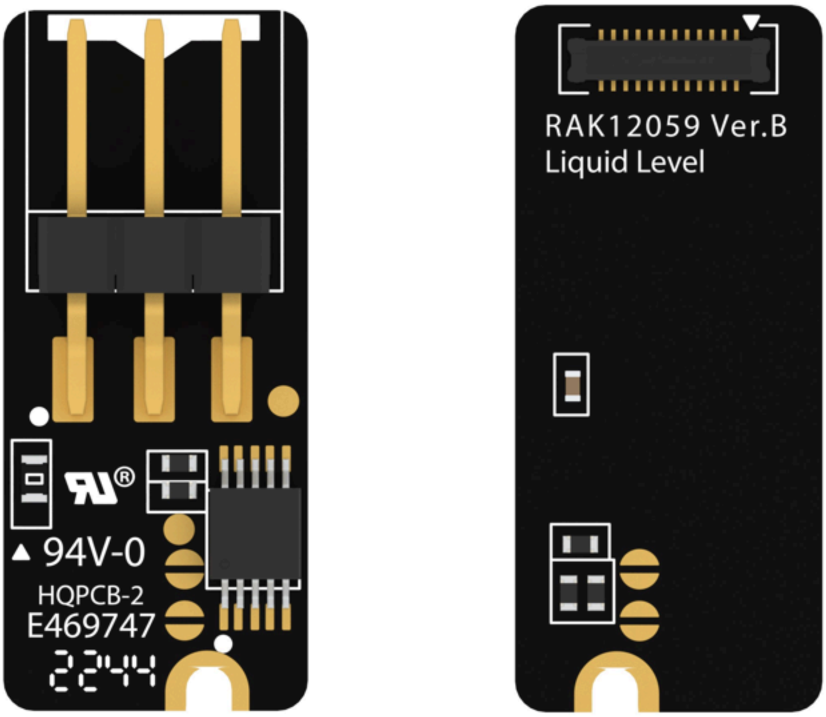

Figure 1: RAK12059 WisBlock Liquid Level Sensor

Description

RAK12059 is a WisBlock Liquid Level Sensor module that extends the WisBlock system with a Liquid Level Sensor from MILONE. It comes with a ready-to-use software library and tutorial, making it easy to build a liquid-level data acquisition system. Liquid-level data is interfaced via I2C. Additionally, it can be mounted to the sensor slot of the WisBlock Base board.

Features

- Sensor specifications

- Active sensor length of MILONE eTape:
  - 213 mm (12110215TC-8TJ)
  - 315 mm (12110215TC-12TJ)
  - 618 mm (12110215TC-24TJ)
- I2C interface
- 3.3 V power supply
- eTape operating temperature range: -9 °C to +65 °C

- **Size**
  - 10 × 23 mm

Specifications

Overview

Mounting

Figure 2 shows the mounting mechanism of the RAK12059 module on a WisBlock Base board. The RAK12059 module can be mounted on the slots: **A, C, D, E, & F**.

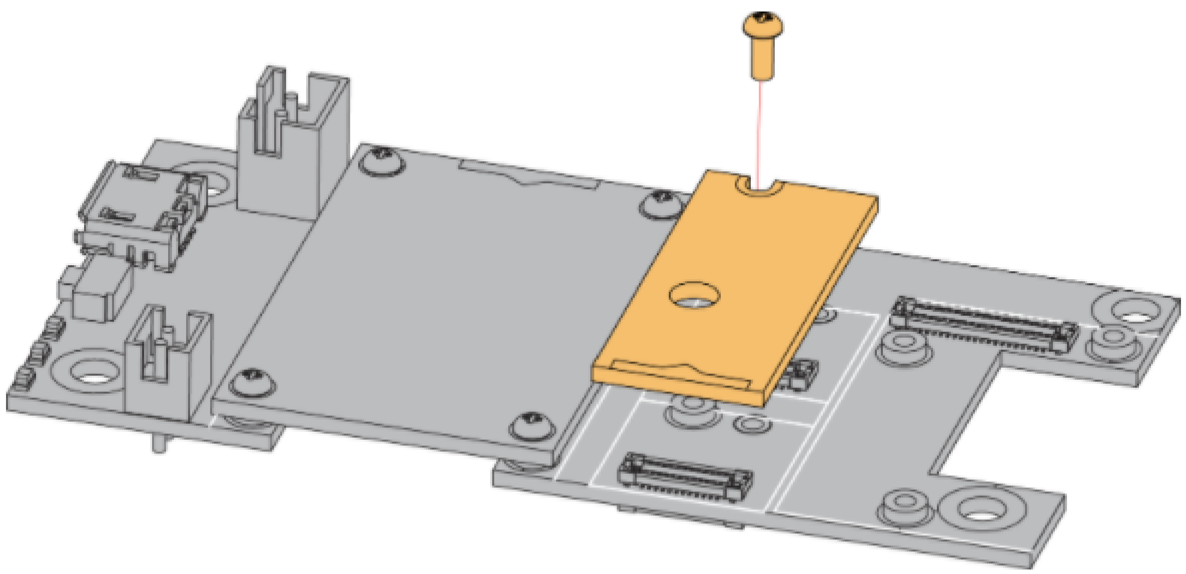

Figure 2: RAK12059 WisBlock Liquid Level Sensor Mounting

Hardware

The hardware specification is categorized into five parts. It shows the module and discusses the pinouts, sensors, and the corresponding functions and diagrams. It also covers the electrical and mechanical parameters, including the tabular data of the functionalities and standard values of the RAK12059 WisBlock Light Sensor.

Chipset

Vendor	Part Number
MILONE	Liquid Level Sensor

Pin Definition

The RAK12059 WisBlock Liquid Level Sensor comprises a standard WisBlock connector. The WisBlock connector allows the RAK12059 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition are shown in **Figure 3**.

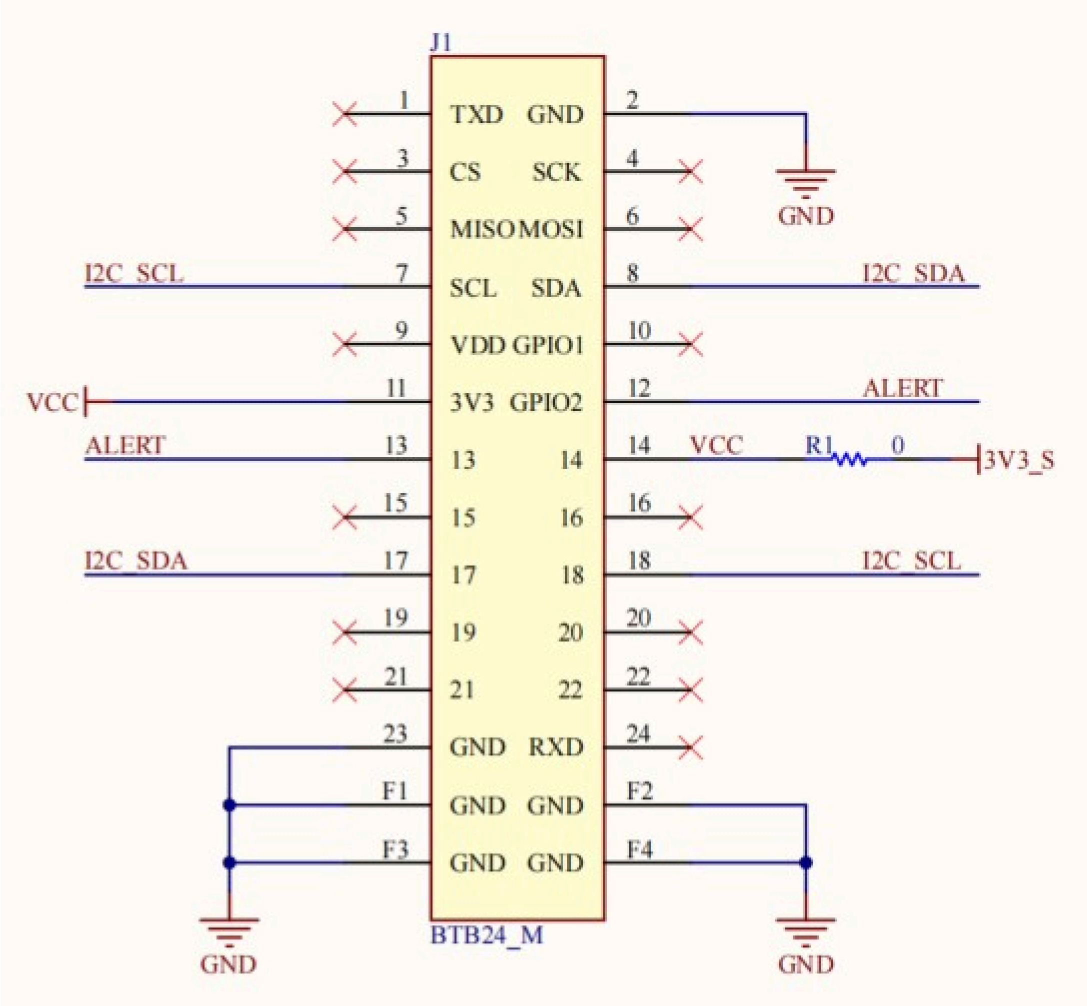

Figure 3: RAK12059 WisBlock Liquid Level Sensor Pinout Diagram

NOTE

- **I2C**-related pins, **ALERT**, **3V3_S**, and **GND** are connected to the WisBlock connector.
- **RAK12059** should not be plugged in **Slot B** as it is dedicated for **IO2 (WisBlock IO2 pin)** that controls the **3V3_S** voltage output.

The **WisBlock Sensor** connector is used to this module and the IO used for **ALERT** pin at **Pin 12** will depend on where sensor slot the module is plugged in. The table shows the compatible pins used by different sensor slots:

ALERT Pin

Slot A	Slot C	Slot D	Slot E	Slot F
IO1	IO3	IO5	IO4	IO6

Electrical Characteristics

Recommended Operating Conditions

Symbol	Description	Min.	Nom.	Max.	Unit
3V3	Power Supply Voltage	-	3.3	-	V

Mechanical Characteristics

Board Dimensions

Figure 4 shows the dimensions and the mechanic drawing of the RAK12059 module.

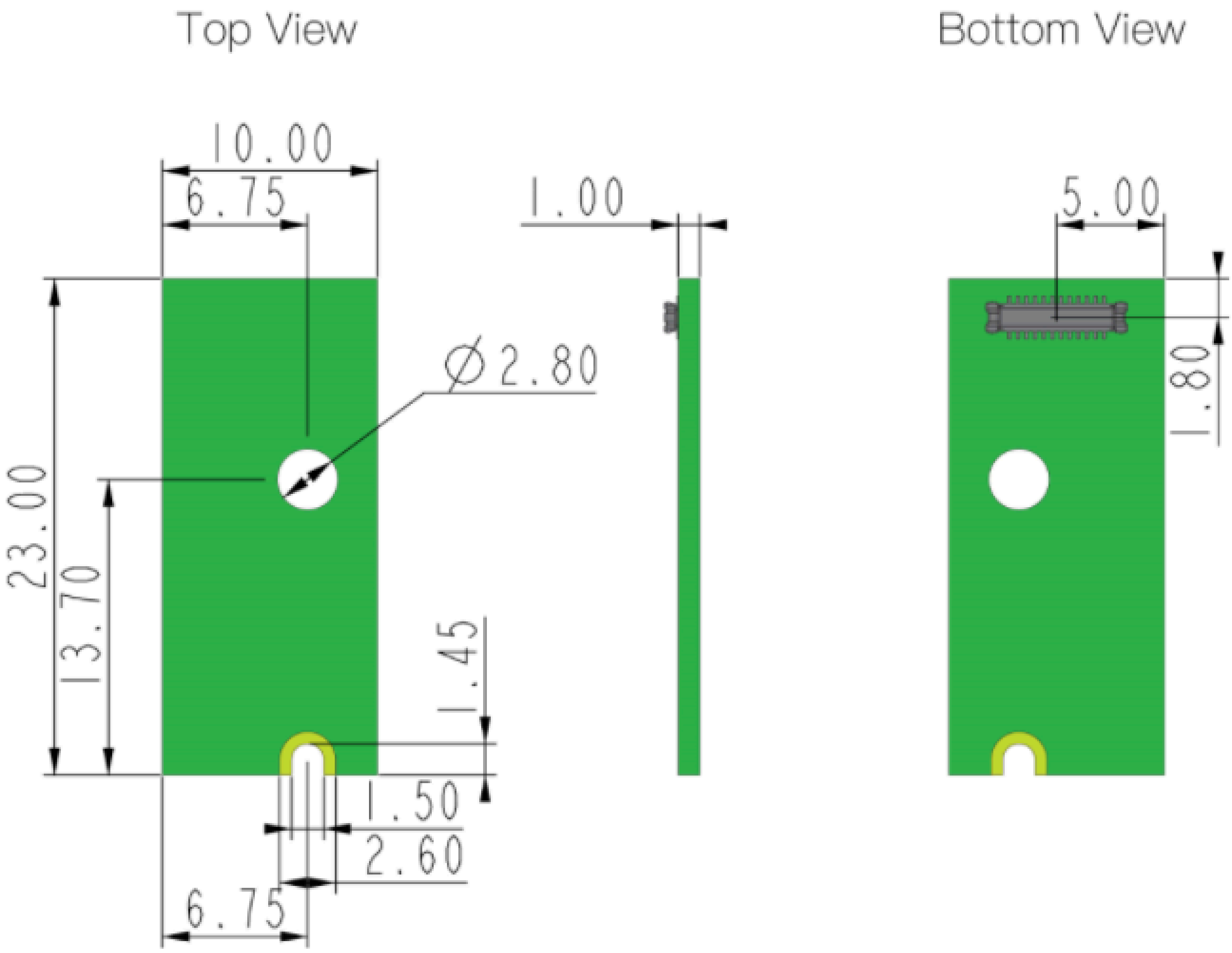

Figure 4: RAK12059 WisBlock Liquid Level Sensor Mechanic Drawing

WisConnector PCB Layout

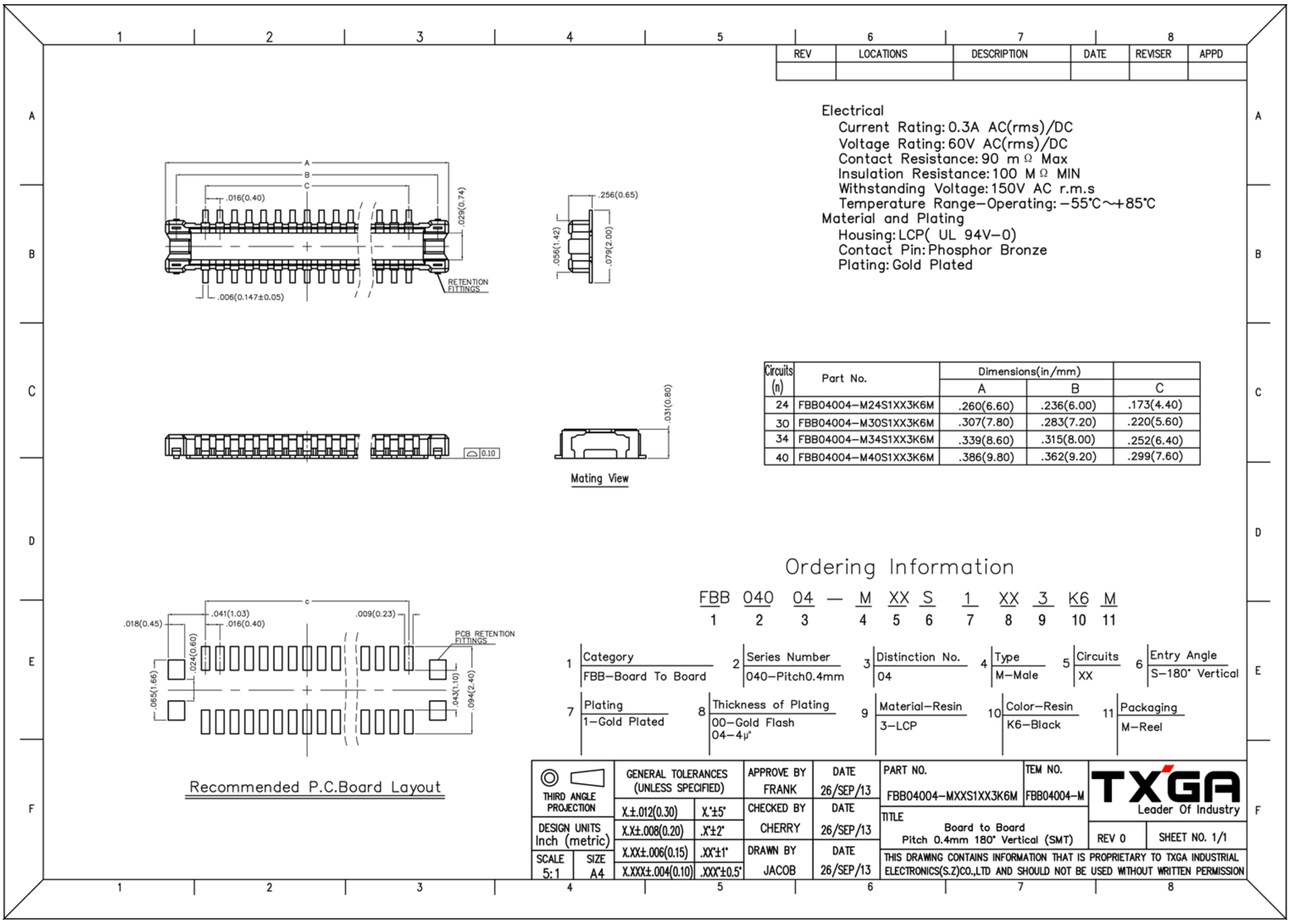

Figure 5: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 6 shows the schematic of the RAK12059 module.

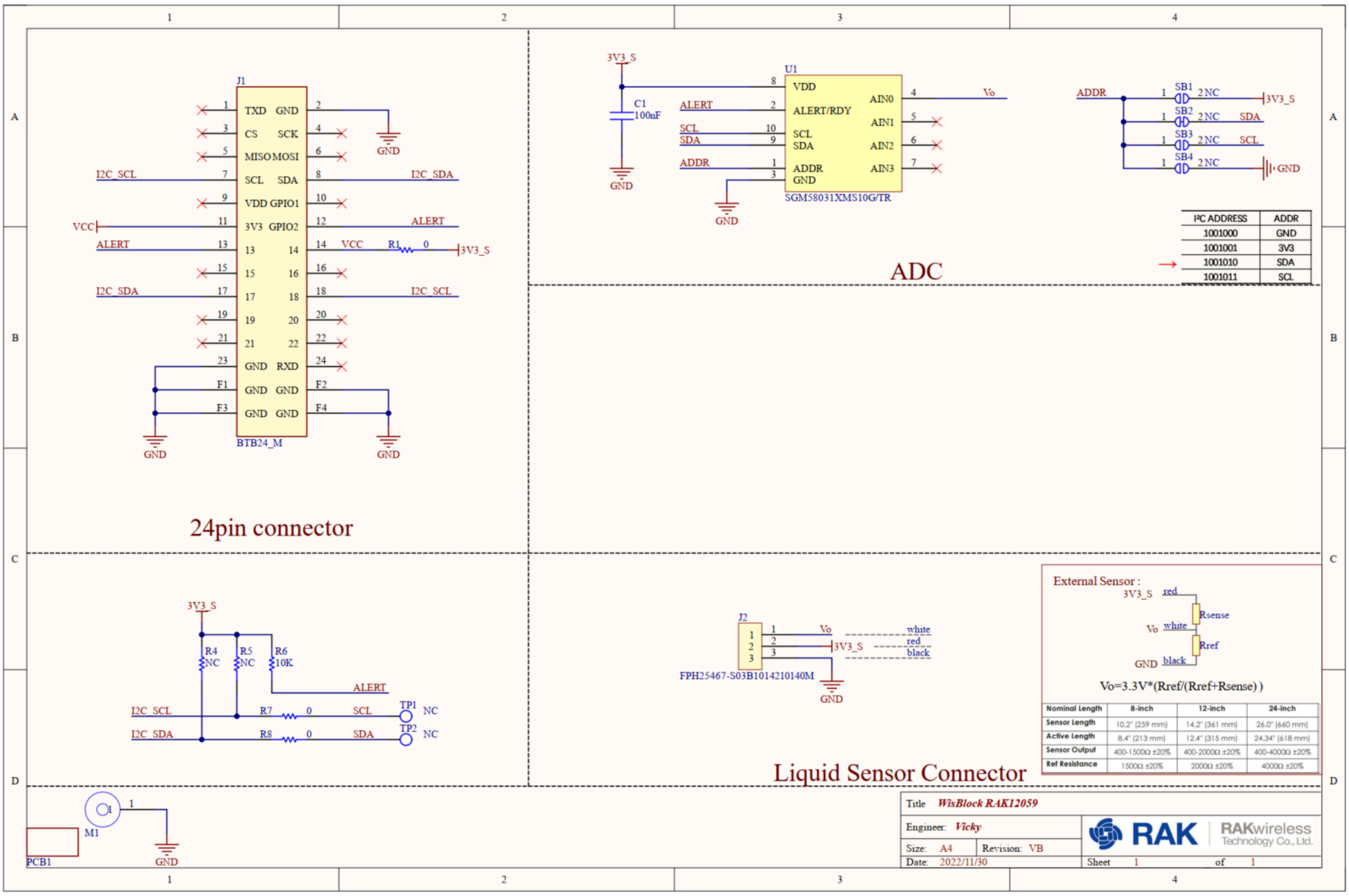

Figure 6: RAK12059 WisBlock Liquid Level Sensor schematics